Starting with MeXpose

```
https://github.com/KoellenspergerLab/MeXpose
```

First-time setup

This beginner-friendly guide describes how to set up the conda environment required to run the MeXpose notebooks.

What we recommend:

- Use **Anaconda/Miniconda with conda** for the most familiar beginner workflow
- Use **mamba** only if you specifically prefer a faster package solver and are already familiar with it.

> **Optional reading:**
> Robert Sandor **"Getting started with Python environments (using Conda)"**:
> https://medium.com/data-science/getting-started-with-python-environments-using-conda-32e9f2779307
> More detailed explanations available on geeksforgeeks.org, **"Getting Started with Conda"**:
> https://www.geeksforgeeks.org/python/what-is-conda/

The MeXpose repository itself provides a ready-made environment file called `mexpose_v2.yaml`.

The repo includes:

- `mexpose_v2.yaml` for creating the conda environment
- `jupyter_lab_config.py` for increasing the JupyterLab upload limit
- notebooks such as `image_reconstruction_TOFWerk.ipynb`, `image_reconstruction_Nu.ipynb`, `data_extraction.ipynb`, `gating.ipynb`, `singlecell_analysis.ipynb`, `quantification_notebook.ipynb`, and `limit_of_detection.ipynb`

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0. Install Anaconda

https://www.anaconda.com/docs/getting-started/anaconda/install/overview

For Windows as well as macOS, graphical installers are available. Run the installer and follow the prompts to complete the setup.

On Windows, use Anaconda Prompt, which is a terminal/command prompt already configured for conda. On macOS/Linux, use the standard Terminal.

The following steps should be performed in the terminal.

1. Get the repository

Download the repository from GitHub or clone it with git. Then open a terminal in the repository folder.

```
git clone https://github.com/KoellenspergerLab/MeXpose.git
cd MeXpose
```

Make sure that your terminal is located in the folder that contains the file `mexpose_v2.yaml`.

2. Create the conda environment

Create the environment using the provided YAML file:

```
conda env create -f mexpose_v2.yaml
```

This may take several minutes, depending on your system and internet connection.

3. Activate the environment

```
conda activate mexpose_v2
```

If activation does not work

First check whether the environment was created correctly:

```
conda env list
```

You should see an environment called `mexpose_v2`.

If `conda activate` is not recognized, conda may not be initialized for your current shell. This is uncommon in Anaconda Prompts, but it may happen in other terminals such as Command Prompt, PowerShell, or the standard macOS/Linux Terminal.

In that case, run:

```
conda init
```

Then close and reopen the terminal, go back into the repository folder, and run:

```
conda activate mexpose_v2
```

Conda documents that `conda init` writes the shell setup needed for activation.

If activation still does not work, check that:

- the environment was created successfully,
- the environment name is correct,
- you are using Anaconda Prompt on Windows (if you are following our instructions here),
- your conda installation is up to date.

4. Start JupyterLab

Once the environment is activated, start JupyterLab from the MeXpose repository folder.

The standard JupyterLab command is:

```
jupyter lab
```

For most users, this step can be skipped at the beginning:
Users can use the repository's large-upload configuration file if necessary, since the notebooks may otherwise hit the default upload limit for files over 10 MB.
Easiest option
Run JupyterLab using the config file that is already in the repo:

```
jupyter lab --config=jupyter_lab_config.py
```

That raises the upload limit to **1 GB**.

5. What you will see in JupyterLab

JupyterLab contains:

- a **left sidebar** with the file browser
- a **main work area** with tabs
- an interface similar to a lightweight development environment.

To open and run a notebook:

1. Open the `notebooks` folder
2. Double-click the notebook you want
3. If prompted, select the kernel, associated with the `mexpose_v2` environment (should not be necessary).
4. Run the cells from top to bottom.

6. Recommended notebook rder

- **TOFWerk raw HDF5 data** → `image_reconstruction_TOFWerk.ipynb`
- **Nu Instruments preprocessed CSV data** → `image_reconstruction_Nu.ipynb`
- Then continue with:
    - `data_extraction.ipynb`
    - `singlecell_analysis.ipynb`
    - `gating.ipynb`
    - `quantification_notebook.ipynb`
    - `limit_of_detection.ipynb`

What file types MeXpose supports

- **TOFWerk** workflows using raw combined laser and TOF data in **HDF5**
- **Nu Instruments** workflows using preprocessed combined laser and TOF data in **CSV**

Other supported file types listed in the pipeline include:

- HDF5
- CSV
- TIFF / OME.TIFF / BIGTIFF
- PNG
- JPEG

Additional formats also work through `imageio`.

Segmentation

For several visualizations and for single-cell extraction, a **greyscale segmentation mask** is required. We recommend **Cellpose-SAM** as the segmentation tool.

So even if the environment and notebooks open correctly, you still need segmentation masks prepared before some analysis steps, as described in our **README files**.

Common mistakes

- **Running `jupyter lab` before activating the environment**
- **Opening the notebook from the wrong folder**

  Start JupyterLab from the MeXpose repository. This makes it easier to access the `notebooks` folder.
- **`conda activate` not recognized**

  Run `conda init`, reopen the terminal, and activate again.
